

EDEXCEL INTERNATIONAL A LEVEL

WMA12/01 October 2024 Worked Solutions

October 2024 paper rebuilt with the worked solution after each question.

11

paper questions

WMA12

Pure 2

**Single-
paper
solutions**standalone IAL
route**Dr Eslam Ahmed**Prepared for Dr Eslam Ahmed - eliteigcse.com**P2**

PAST PAPER

WMA12/01 October 2024

October 2024 | 11 questions | 75 marks

11

questions

75

marks

Answers

worked solution
after each
question

Question 1

Integration

1. A continuous curve has equation $y = f(x)$.

A table of values of x and y for $y = f(x)$ is shown below.

x	0.5	1.75	3	4.25	5.5
y	3.479	6.101	7.448	6.823	5.182

Using the trapezium rule with all the values of y in the given table,

- (a) find an estimate for

$$\int_{0.5}^{5.5} f(x) \, dx$$

giving your answer to one decimal place.

(3)

- (b) Using your answer to part (a) and making your method clear, estimate

$$\int_{0.5}^{5.5} (f(x) + 4x) \, dx$$

(2)

Worked Solution - Question 1

1. Find the strip width

The x values increase by **1.25**, so the trapezium width is $h = 1.25$.

2. Apply the trapezium rule

$$\int_{0.5}^{5.5} f(x) dx \approx \frac{1.25}{2} \{3.479 + 5.182 + 2(6.101 + 7.448 + 6.823)\}.$$

3. Evaluate the estimate

This gives **30.878125**, which is **30.9** to one decimal place.

4. Split the second integral

$$\int_{0.5}^{5.5} (f(x) + 4x) dx = \int_{0.5}^{5.5} f(x) dx + \int_{0.5}^{5.5} 4x dx.$$

5. Integrate $4x$

$$\int_{0.5}^{5.5} 4x dx = [2x^2]_{0.5}^{5.5} = 2(5.5)^2 - 2(0.5)^2 = 60.$$

6. Use part a

Using the estimate from part (a), the required value is **30.9 + 60 = 90.9**.

Final answer

(a) 30.9.

(b) 90.9.

Question 2

Sequences & Series

2. A sequence of numbers $u_1, u_2, u_3, \dots$ is defined by

$$u_1 = 7$$

$$u_{n+1} = (-1)^n u_n + k$$

where k is a constant.

- (a) Show that $u_5 = 7$

(3)

Given that $\sum_{r=1}^4 u_r = 30$

- (b) find the value of k .

(2)

- (c) Hence find the value of $\sum_{r=1}^{150} u_r$

(2)

Worked Solution - Question 2

Topic group

1. Find the first few terms

$$u_1 = 7, u_2 = (-1)^1 u_1 + k = k - 7, u_3 = (-1)^2 u_2 + k = 2k - 7, \text{ and} \\ u_4 = (-1)^3 u_3 + k = 7 - k.$$

2. Show u five

$$u_5 = (-1)^4 u_4 + k = (7 - k) + k = 7.$$

3. Use the given sum

$$\sum_{r=1}^4 u_r = 7 + (k - 7) + (2k - 7) + (7 - k) = 2k.$$

4. Find k

$$\text{Since } \sum_{r=1}^4 u_r = 30, \text{ we have } 2k = 30, \text{ so } k = 15.$$

5. Write one repeated block

With $k = 15$, the first four terms are **7, 8, 23, -8**, and their sum is **30**. Since $u_5 = u_1$, the pattern repeats every four terms.

6. Sum 150 terms

There are **37** complete blocks in the first **150** terms, with the first two terms left over. Hence $\sum_{r=1}^{150} u_r = 37(30) + 7 + 8 = 1125$.

Final answer

(a) Proven.

(b) $k = 15$.

(c) 1125.

Question 3

Polynomials

3.

$$f(x) = 2x^3 - x^2 + Ax + B$$

where A and B are integers.

Given that when $f(x)$ is divided by $(x + 3)$ the remainder is 55

(a) show that

$$3A - B = -118$$

(2)

Given also that $(2x - 5)$ is a factor of $f(x)$,

(b) find the value of A and the value of B .

(3)

(c) Hence find the quotient when $f(x)$ is divided by $(x - 7)$

(2)

Worked Solution - Question 3

1. Use the remainder theorem

Dividing by $x + 3$ leaves remainder 55, so $f(-3) = 55$.

2. Show the given equation

$f(-3) = 2(-3)^3 - (-3)^2 - 3A + B = -63 - 3A + B$. Therefore $-63 - 3A + B = 55$, so $3A - B = -118$.

3. Use the factor theorem

Since $2x - 5$ is a factor, $x = \frac{5}{2}$ is a root and $f\left(\frac{5}{2}\right) = 0$.

4. Form a second equation

$2\left(\frac{5}{2}\right)^3 - \left(\frac{5}{2}\right)^2 + \frac{5}{2}A + B = 0$, so $25 + \frac{5}{2}A + B = 0$. Multiplying by 2 gives $5A + 2B = -50$.

5. Solve simultaneously

From $3A - B = -118$, $B = 3A + 118$. Substitute into $5A + 2B = -50$ to get $11A = -286$, so $A = -26$ and $B = 40$.

6. Write the polynomial

$$f(x) = 2x^3 - x^2 - 26x + 40.$$

7. Divide by x minus seven

$2x^3 - x^2 - 26x + 40 = (x - 7)(2x^2 + 13x + 65) + 495$, so the quotient is $2x^2 + 13x + 65$.

Final answer

(a) Proven.

(b) $A = -26, B = 40$.

(c) $2x^2 + 13x + 65$.

Question 4

Applications of Differentiation

4. In this question you must show all stages of your working.

Solutions relying entirely on calculator technology are not acceptable.

The curve C has equation

$$y = 4x^{\frac{1}{2}} + 9x^{-\frac{1}{2}} + 3 \quad x > 0$$

- (a) Find $\frac{dy}{dx}$ giving each term in simplest form. (2)
- (b) Hence find the x coordinate of the stationary point of C . (2)
- (c) (i) Find $\frac{d^2y}{dx^2}$ giving each term in simplest form.
- (ii) Hence determine the nature of the stationary point of C , giving a reason for your answer. (2)
- (d) State the range of values of x for which y is decreasing. (1)

Worked Solution - Question 4

Topic group

1. Rewrite using powers

$$y = 4x^{1/2} + 9x^{-1/2} + 3.$$

2. Differentiate

$$\frac{dy}{dx} = 2x^{-1/2} - \frac{9}{2}x^{-3/2} = \frac{2}{\sqrt{x}} - \frac{9}{2x^{3/2}}.$$

3. Set the derivative to zero

$$\frac{dy}{dx} = \frac{4x - 9}{2x^{3/2}}. \text{ Since } x > 0, \text{ the stationary point occurs when } 4x - 9 = 0.$$

4. Find the x coordinate

$$\text{Therefore } x = \frac{9}{4}.$$

5. Differentiate again

$$\frac{d^2y}{dx^2} = -x^{-3/2} + \frac{27}{4}x^{-5/2} = -\frac{1}{x^{3/2}} + \frac{27}{4x^{5/2}}.$$

6. Test the stationary point

$$\frac{d^2y}{dx^2} = \frac{27 - 4x}{4x^{5/2}}. \text{ At } x = \frac{9}{4}, \text{ the numerator is } 27 - 9 = 18 > 0, \text{ so the stationary point is a minimum.}$$

7. Find where y is decreasing

$$\text{For } x > 0, 2x^{3/2} > 0, \text{ so } \frac{dy}{dx} < 0 \text{ when } 4x - 9 < 0. \text{ Hence } y \text{ is decreasing for } 0 < x < \frac{9}{4}.$$

Final answer

$$(a) \frac{2}{\sqrt{x}} - \frac{9}{2x^{3/2}}.$$

$$(b) x = \frac{9}{4}.$$

$$(c)(i) -\frac{1}{x^{3/2}} + \frac{27}{4x^{5/2}}.$$

(c)(ii) Minimum.

$$(d) 0 < x < \frac{9}{4}.$$

Question 5

Binomial Expansion

5. (a) Find, in terms of a , the first 3 terms, in ascending powers of x , of the binomial expansion of

$$(2 + ax)^6$$

where a is a non-zero constant. Give each term in simplest form.

(3)

$$f(x) = \left(3 + \frac{1}{x}\right)^2 (2 + ax)^6$$

Given that the constant term in the expansion of $f(x)$ is 576

- (b) find the value of a .

(4)

Worked Solution - Question 5

1. Write the first three binomial terms

$$(2 + ax)^6 = 2^6 + 6(2^5)(ax) + \binom{6}{2}(2^4)(ax)^2 + \dots$$

2. Simplify the terms

The first three terms are $64 + 192ax + 240a^2x^2$.

3. Expand the other factor

$$\left(3 + \frac{1}{x}\right)^2 = 9 + 6x^{-1} + x^{-2}.$$

4. Collect constant contributions

The constant term in $\left(3 + \frac{1}{x}\right)^2 (2 + ax)^6$ is $9(64) + 6(192a) + 240a^2$.

5. Use the given constant term

$$576 + 1152a + 240a^2 = 576, \text{ so } 240a^2 + 1152a = 0.$$

6. Solve for a

$$48a(5a + 24) = 0. \text{ Since } a \neq 0, 5a + 24 = 0, \text{ so } a = -\frac{24}{5}.$$

Final answer

$$(a) 64 + 192ax + 240a^2x^2.$$

$$(b) a = -\frac{24}{5}.$$

Question 6

Laws of Logarithms

6.

In this question you must show all stages of your working.

Solutions relying entirely on calculator technology are not acceptable.

Using the laws of logarithms, solve

$$\log_4(12 - 2x) = 2 + 2\log_4(x + 1)$$

(5)

Worked Solution - Question 6

Topic group

1. State the domain

The logarithms require $12 - 2x > 0$ and $x + 1 > 0$, so $-1 < x < 6$.

2. Combine the right hand side

$$2 = \log_4(16), \text{ so } 2 + 2\log_4(x + 1) = \log_4(16(x + 1)^2).$$

3. Equate the arguments

$$\log_4(12 - 2x) = \log_4(16(x + 1)^2) \text{ gives } 12 - 2x = 16(x + 1)^2.$$

4. Form the quadratic

$$12 - 2x = 16x^2 + 32x + 16, \text{ so } 8x^2 + 17x + 2 = 0.$$

5. Factor and solve

$$8x^2 + 17x + 2 = (8x + 1)(x + 2) = 0, \text{ so } x = -\frac{1}{8} \text{ or } x = -2.$$

6. Check the domain

The value $x = -2$ is outside $-1 < x < 6$, so the solution is $x = -\frac{1}{8}$.

Final answer

$$x = -\frac{1}{8}.$$

Question 7

Modelling with Sequences & Series

7. Jem pays money into a savings scheme, A , over a period of 300 months.

Jem pays £20 into scheme A in month 1, £20.50 in month 2, £21 in month 3 and so on, so that the amounts Jem pays each month form an arithmetic sequence.

- (a) Show that Jem pays £69.50 into scheme A in month 100 (1)

- (b) Find the **total** amount that Jem pays into scheme A over the period of 300 months. (2)

Kim pays money into a different savings scheme, B , over the same period of 300 months.

In a model, the amounts Kim pays into scheme B increase by the same percentage each month, so that the amounts Kim pays each month form a geometric sequence.

Given that Kim pays

- £20 into scheme B in month 1
- £250 into scheme B in month 300

- (c) use the model to calculate, to the nearest £10, the difference between the total amount paid into scheme A and the total amount paid into scheme B over the period of 300 months. (3)

Worked Solution - Question 7

Topic group

1. Identify Jem's sequence

Jem's payments are arithmetic with first term **20** and common difference **0.50**.

2. Show month 100

The month **100** payment is $20 + 99(0.50) = 69.50$, so Jem pays £69.50.

3. Sum Jem's payments

$$S_{300} = \frac{300}{2} \{2(20) + 299(0.50)\} = 150(189.5) = 28425.$$

4. Model Kim's payments

Let the common ratio be r . Since the first payment is **20** and the month **300** payment is **250**, $20r^{299} = 250$.

5. Find the ratio

Thus $r^{299} = 12.5$ and $r = 12.5^{1/299}$.

6. Sum Kim's payments

$$S_B = 20 \frac{r^{300} - 1}{r - 1} \text{ with } r = 12.5^{1/299}, \text{ giving } S_B \approx 27362.95.$$

7. Find the difference

The difference is $28425 - 27362.95 \approx 1062.05$, which is £1,060 to the nearest £10.

Final answer

(a) Proven.

(b) £28,425.

(c) £1,060.

Question 8

Integration

8.

In this question you must show all stages of your working.

Solutions relying entirely on calculator technology are not acceptable.

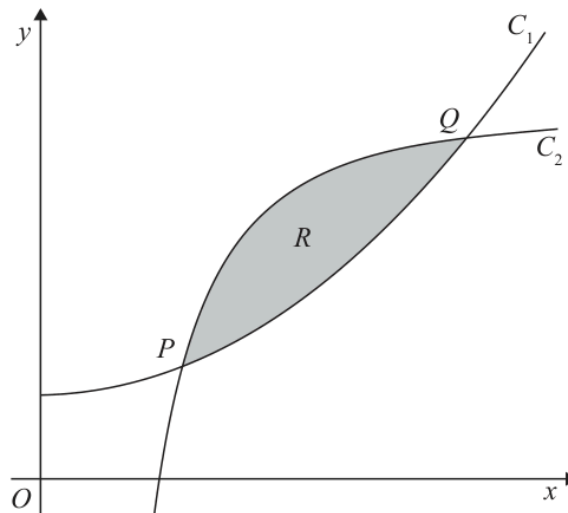


Figure 1

Figure 1 shows a sketch of part of the curve C_1 with equation

$$y = x^2 + 3 \quad x > 0$$

and part of the curve C_2 with equation

$$y = 13 - \frac{9}{x^2} \quad x > 0$$

The curves C_1 and C_2 intersect at the points P and Q as shown in Figure 1.(a) Use algebra to find the x coordinate of P and the x coordinate of Q .

(4)

The finite region R , shown shaded in Figure 1, is bounded by C_1 and C_2 (b) Use algebraic integration to find the exact area of R .

(4)

Worked Solution - Question 8

1. Equate the curves

$$x^2 + 3 = 13 - \frac{9}{x^2}.$$

2. Clear the denominator

Multiplying by x^2 gives $x^4 + 3x^2 = 13x^2 - 9$, so $x^4 - 10x^2 + 9 = 0$.

3. Factor

Let $u = x^2$. Then $u^2 - 10u + 9 = (u - 1)(u - 9) = 0$, so $x^2 = 1$ or $x^2 = 9$.

4. Use x greater than zero

Since $x > 0$, the intersections have $x = 1$ and $x = 3$. Thus $x_P = 1$ and $x_Q = 3$.

5. Set up the area

Between $x = 1$ and $x = 3$, curve C_2 is above curve C_1 , so

$$R = \int_1^3 \left(13 - \frac{9}{x^2} - (x^2 + 3) \right) dx.$$

6. Simplify the integrand

$$R = \int_1^3 (10 - x^2 - 9x^{-2}) dx.$$

7. Integrate

$$R = \left[10x - \frac{x^3}{3} + \frac{9}{x} \right]_1^3.$$

8. Evaluate

$$R = (30 - 9 + 3) - \left(10 - \frac{1}{3} + 9 \right) = 24 - \frac{56}{3} = \frac{16}{3}.$$

Final answer

(a) $x_P = 1, x_Q = 3$.

(b) $\frac{16}{3}$.

Question 9

Trigonometric Equations

9.

In this question you must show detailed reasoning.**Solutions relying entirely on calculator technology are not acceptable.**

(a) Show that the equation

$$2 \tan \theta = 3 \cos \theta$$

can be written as

$$3 \sin^2 \theta + 2 \sin \theta - 3 = 0 \quad (3)$$

(b) Hence solve, for $-\pi < x < \pi$, the equation

$$2 \tan \left(2x + \frac{\pi}{3} \right) = 3 \cos \left(2x + \frac{\pi}{3} \right)$$

giving your answers to 3 significant figures.

(4)

Worked Solution - Question 9

1. Write tan in terms of sin and cos

$$2 \tan \theta = 3 \cos \theta \text{ gives } 2 \frac{\sin \theta}{\cos \theta} = 3 \cos \theta.$$

2. Multiply by cos theta

Since $\tan \theta$ is defined, $\cos \theta \neq 0$. Hence $2 \sin \theta = 3 \cos^2 \theta$.

3. Use the identity

$$2 \sin \theta = 3(1 - \sin^2 \theta), \text{ so } 3 \sin^2 \theta + 2 \sin \theta - 3 = 0.$$

4. Substitute the new angle

$$\text{Let } \theta = 2x + \frac{\pi}{3}. \text{ Then } 3 \sin^2 \theta + 2 \sin \theta - 3 = 0.$$

5. Solve for sin theta

$$\sin \theta = \frac{-2 \pm \sqrt{40}}{6} = \frac{-1 \pm \sqrt{10}}{3}. \text{ The root } \frac{-1 - \sqrt{10}}{3} \text{ is less than } -1, \text{ so reject it.}$$

6. Find the angle range

$$\text{Since } -\pi < x < \pi, -\frac{5\pi}{3} < \theta < \frac{7\pi}{3}.$$

7. Find the theta values

$$\text{For } \sin \theta = \frac{-1 + \sqrt{10}}{3} \text{ in this range, } \theta \approx -3.94649, 0.804897, 2.33670, \text{ and } 7.08808.$$

8. Convert back to x

$$\text{Using } x = \frac{\theta - \pi/3}{2} \text{ gives } x \approx -2.49684, -0.121150, 0.644749, \text{ and } 3.02044.$$

9. Round the answers

To 3 significant figures, $x = -2.50, -0.121, 0.645$, and 3.02 .

Final answer

(a) Proven.

(b) $x = -2.50, -0.121, 0.645, 3.02$.

Question 10

Circles

10. The circle C has equation

$$x^2 + y^2 + 4x - 30y + 209 = 0$$

(a) Find

- (i) the coordinates of the centre of C ,
- (ii) the exact value of the radius of C .

(3)

The line L has equation $y = mx + 1$, where m is a constant.

Given that L is the tangent to C at the point P ,

(b) show that

$$2m^2 - 7m - 22 = 0$$

(3)

(c) Hence find the possible pairs of coordinates of P .

(4)

Worked Solution - Question 10

Topic group

1. Complete the square $x^2 + y^2 + 4x - 30y + 209 = 0$ becomes $(x + 2)^2 + (y - 15)^2 = 20$.**2. Read the centre and radius**The centre is $(-2, 15)$ and the radius is $\sqrt{20} = 2\sqrt{5}$.**3. Use the tangent distance**The line $y = mx + 1$ can be written as $mx - y + 1 = 0$. Its distance from $(-2, 15)$ must be $2\sqrt{5}$.**4. Set up the equation for m**

$$\frac{|m(-2) - 15 + 1|}{\sqrt{m^2 + 1}} = 2\sqrt{5}, \text{ so } \frac{|-2m - 14|}{\sqrt{m^2 + 1}} = 2\sqrt{5}.$$

5. Show the quadraticSquaring gives $4(m + 7)^2 = 20(m^2 + 1)$. Hence $(m + 7)^2 = 5(m^2 + 1)$, so $2m^2 - 7m - 22 = 0$.**6. Find the possible gradients**

$$2m^2 - 7m - 22 = 0 \text{ gives } m = -2 \text{ or } m = \frac{11}{2}.$$

7. Use perpendicular radii

A tangent is perpendicular to the radius at the point of contact.

8. Find the first point

For $m = -2$, the radius gradient is $\frac{1}{2}$. The radius through the centre is $y - 15 = \frac{1}{2}(x + 2)$. Solving with $y = -2x + 1$ gives $x = -6$ and $y = 13$.

9. Find the second point

For $m = \frac{11}{2}$, the radius gradient is $-\frac{2}{11}$. The radius through the centre is $y - 15 = -\frac{2}{11}(x + 2)$. Solving with $y = \frac{11}{2}x + 1$ gives $x = \frac{12}{5}$ and $y = \frac{71}{5}$.

Final answer

(a)(i) $(-2, 15)$.

(a)(ii) $2\sqrt{5}$.

(b) Proven.

(c) $(-6, 13)$ and $\left(\frac{12}{5}, \frac{71}{5}\right)$.

Question 11

Proof

11. (i) Prove by counter example that the statement

“If n is a prime number then $3^n + 2$ is also a prime number.”

is false.

(2)

- (ii) Use proof by exhaustion to prove that if m is an integer that is **not** divisible by 3, then

$$m^2 - 1$$

is divisible by 3

(4)

Worked Solution - Question 11

Topic group

1. Choose a prime value

Take $n = 5$, which is prime.

2. Evaluate the expression

$3^5 + 2 = 243 + 2 = 245 = 5 \times 49$, so it is not prime. This disproves the statement.

3. List the possible forms of m

If integer m is not divisible by 3, then m has remainder 1 or 2 on division by 3.

4. Case one

If $m = 3q + 1$, then $m^2 - 1 = (3q + 1)^2 - 1 = 9q^2 + 6q = 3(3q^2 + 2q)$, which is divisible by 3.

5. Case two

If $m = 3q + 2$, then

$m^2 - 1 = (3q + 2)^2 - 1 = 9q^2 + 12q + 3 = 3(3q^2 + 4q + 1)$, which is divisible by 3.

6. Conclude by exhaustion

Both possible cases for an integer not divisible by 3 have been checked, so $m^2 - 1$ is divisible by 3.

Final answer

(i) Counterexample: $n = 5$ gives $3^5 + 2 = 245$, not prime.

(ii) Proven.